

Chapter 4 Critical Thinking Questions (page 92: #1 - #6 and #8):

1.

- a) if (grade >= 90) {
 System.out.println("Great job!");
}
- b) if (number < 20 || number > 50) {
 System.out.println("Error");
}
- c) if (y < 100) {
 y = y + 2;
}

2.

```
if (num1 >= num2) {  
    System.out.println("First number is larger");  
}  
else if (num1 <= num2) {  
    System.out.println("Second number is larger");  
}  
else if (num1 == num2) {  
    System.out.println("Numbers are equal");  
}
```

3.

Blank1 = even
Blank2 = odd

4.

- a) int number = (int) (Math.random() * 50) + 1;
- b) int number = (int) (Math.random() * 81) + 20;
- c) double number = (Math.random() * 10) + 10;

5.

The else if statement for adults does not need to include "age > 18" because that part only runs if it has not already been identified that the person is a child. Additionally, the last else if statement should just be an "else" statement because it is that last possible case, this also means you don't need to specify (age > 65).

6.

- a) True
- b) False
- c) True
- d) True
- e) True
- f) True
- g) True

8.

- a) True
- b) False, an if statement nested inside another if only runs if the first statement is true, whereas an if else statement only runs if the first statement doesn't (it's an alternative).
- c) False, it cannot be a double because the values couldn't be compared and would not be exact matches.
- d) True
- e) False, you need the (int) cast instead for integers.
- f) True
- g) True
- h) False, ! is actually evaluated first and would only apply to the entire expression if parentheses are used.
- i) True
- j) False, it would usually require the class name "Math" in front.